Mass Gap for the O(N) Model: Analysis of the HS Approach

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Abstract

We analyze the Hubbard-Stratonovich approach to proving the mass gap for the O(N) model in two dimensions at large N. The HS identity for the Euclidean quartic requires imaginary coupling, creating a sign problem. We compare the NLSM (where Kupiainen's constraint trick gives a real mass m_0^2 in the operator, enabling a direct FK bound for each σ-value) with the LSM (where the real mass cancels after HS, and the mass gap must emerge from the oscillatory σ-average). A comparison across $d = 0, 1, 2$ shows that the sign problem is specific to $d = 2$: the logarithmic UV divergence of the Green's function promotes the self-energy from $O(1/N)$ to $O(1)$, causing dimensional transmutation and making the oscillatory cancellations essential. We discuss three proof strategies and the current status of the Lean formalization.

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1 The HS identity and the sign problem

1.1 The Euclidean quartic requires imaginary coupling

The O(N) LSM action on $\Lambda \subset \mathbb{Z}^2$:

$$S[\varphi] = \tfrac{1}{2}\sum_i \langle \varphi^i, (-\Delta)\varphi^i\rangle + N\sum_x \lambda\big(\tfrac{|\varphi(x)|^2}{N} - v^2\big)^2. \tag{1}$$

The Boltzmann weight is e^{-S}. The quartic appears as $e^{-\lambda a^2}$ with $a = |\varphi|^2/N - v^2$. The HS identity:

$$e^{-\lambda a^2} = \frac{1}{\sqrt{4\pi\lambda}}\int_{-\infty}^{\infty} e^{-\sigma^2/(4\lambda)\,+\,i\sigma a}\,d\sigma. \tag{2}$$

Here $\sigma \in \mathbb{R}$ (real integration variable), $e^{-\sigma^2/(4\lambda)}$ ensures convergence, and the coupling $i\sigma a$ is **imaginary**. This is a Fourier transform (proved from Mathlib's `fourierIntegral_gaussian`).

The imaginary coupling is forced by the *minus* sign in $e^{-\lambda a^2}$ (repulsive/confining quartic in Euclidean signature). A real coupling would give $e^{+\lambda a^2}$ (attractive, wrong sign).

1.2 The σ-integral on the real axis

After applying (2) at each site and integrating out φ (Gaussian with complex mass), the partition function becomes:

$$Z = c\int_{\mathbb{R}^{|\Lambda|}} \det\big(-\Delta + 2i\sigma z\big)^{-N/2}\, e^{-\sum_x[\sigma(x)^2/(4\lambda)+i\sigma(x)v^2]}\, d^{|\Lambda|}\sigma, \tag{3}$$

where $z = N^{-1/2}$ and the det comes from the φ-integral.

This integral **converges** on the real σ-axis:

- $e^{-\sigma^2/(4\lambda)}$ provides Gaussian decay in σ.
- $|\det(-\Delta + 2i\sigma z)^{-N/2}| \leq \det(-\Delta)^{-N/2}$ (bounded, since $|1 + i\theta| \geq 1$ for real θ).

But the integrand is **complex** (the det and the $e^{i\sigma v^2}$ factor oscillate). This is the sign problem.

2 The NLSM: Kupiainen's mass trick

2.1 Adding m_0^2 for free

For the NLSM with constraint $|\varphi|^2 = N\beta$: any function of $|\varphi|^2$ is constant on the constraint surface. In particular, $\frac{1}{2}m_0^2|\varphi|^2 = \frac{1}{2}m_0^2 N\beta = \text{const}$.

So we can **freely add** $\frac{1}{2}m_0^2|\varphi|^2$ to the action, giving the massive GFF:

$$S_{\text{NLSM}} = \tfrac{1}{2}\langle\varphi, (-\Delta + m_0^2)\varphi\rangle + \text{const}, \qquad |\varphi|^2 = N\beta. \tag{4}$$

The parameter $m_0^2 > 0$ is arbitrary; we choose it via the gap equation (Kupiainen eq. 9b): $(-\Delta + m_0^2)^{-1}_{ii} = \beta$.

2.2 The NLSM operator has real mass

After HS (introducing the Lagrange multiplier α for the constraint), the φ-propagator at fixed α is:

$$\left(-\Delta + m_0^2 - 2i\alpha z\right)^{-1}(x, 0). \tag{5}$$

The operator has **real part** $-\Delta + m_0^2 \geq m_0^2 > 0$. By the triangle inequality (FK representation):

$$\left|(-\Delta + m_0^2 - 2i\alpha z)^{-1}(x, 0)\right| \leq (-\Delta + m_0^2)^{-1}(x, 0) \sim e^{-m_0|x|}. \tag{6}$$

Every α-value gives exponential decay with mass m_0. The α-average is a correction, not the leading effect.

2.3 Kupiainen's proof structure

1. Gap equation: $m_0 > 0$ (always in 2d).
2. Bound (6): each α-value gives decay.
3. Cluster expansion + RP: control the α-average (prefactor, not the mass).
4. Mass gap: $m = m_0(1 + O(1/N))$.

The hard step is (3), but the mass m_0 comes for free from (2).

3 The LSM: the mass cancels

3.1 Add-subtract fails

For the LSM (no constraint), we try the same trick: add $\frac{1}{2}m_0^2|\varphi|^2$ to the action and subtract it from the interaction:

$$S = \left[\tfrac{1}{2}\langle\varphi, (-\Delta + m_0^2)\varphi\rangle\right] + \left[\lambda(u - v^2)^2 - \tfrac{1}{2}m_0^2 u\right], \qquad u = |\varphi|^2/N. \tag{7}$$

Apply HS to $\lambda(u - v^2)^2$ only. The subtracted mass $-\frac{1}{2}m_0^2 u$ remains outside the HS. Computing the φ-dependence:

$$\begin{aligned} &-\tfrac{1}{2}\langle\varphi, (-\Delta + m_0^2)\varphi\rangle + i\sigma|\varphi|^2 + \tfrac{1}{2}m_0^2|\varphi|^2 \\ &\quad = -\tfrac{1}{2}\langle\varphi, (-\Delta)\varphi\rangle + i\sigma|\varphi|^2 \\ &\quad = -\tfrac{1}{2}\langle\varphi, (-\Delta - 2i\sigma)\varphi\rangle. \end{aligned} \tag{8}$$

The m_0^2 from the GFF and the $-m_0^2$ from the subtraction **cancel exactly**. The LSM operator is:

$$-\Delta - 2i\sigma \qquad (\textbf{no real mass}). \tag{9}$$

3.2 Consequences

The operator (9) has eigenvalues $k^2 - 2i\sigma$ with real part $k^2 \geq 0$ (from $-\Delta$) and imaginary part -2σ. For $k = 0$: the eigenvalue is $-2i\sigma$ (purely imaginary).

- **No single-σ decay**: $|(-\Delta - 2i\sigma)^{-1}(x, 0)|$ does NOT decay exponentially for each σ-value. At $k = 0$: $|1/(-2i\sigma)| = 1/(2|\sigma|)$, independent of $|x|$.
- **Mass from σ-average only**: The exponential decay of $\langle\varphi(x)\varphi(0)\rangle$ must come *entirely* from the oscillatory cancellations in the σ-integral (3).
- **The FK bound fails**: The NLSM bound (6) relies on the real mass m_0^2. Without it, $|G_\sigma| \leq G_0$ gives $G_0 = (-\Delta)^{-1}$ (massless propagator, no decay).

3.3 Comparison

	NLSM	LSM
Mass in operator	m_0^2 (free, from constraint)	**none** (cancels)
Operator after HS	$-\Delta + m_0^2 - 2i\alpha z$	$-\Delta - 2i\sigma$
Single-σ decay	$e^{-m_0\|x\|}$ (yes)	**none**
Mass gap source	Real mass + corrections	**σ-average only**
FK bound	Works ($\|G\| \leq G_{m_0}$)	**Fails**

4 Comparison across dimensions

The sign problem and mass gap structure depend strongly on dimension.

4.1 $d = 0$ (matrix model)

One site ($|\Lambda| = 1$). The σ-integral is one-dimensional: $\int_{-\infty}^{\infty} e^{-\sigma^2/(4\lambda)+i\sigma a}\, d\sigma$. The saddle-point Gaussian is exact up to $O(1/N)$ corrections. Oscillations are $O(1/N^2)$ — negligible. No sign problem.

The Green's function $C = 1/m_0^2$ is a number. The self-energy $\Sigma = O(1/N)$. The mass is perturbative: $m = m_{\text{bare}}(1 + O(1/N))$. No dimensional transmutation.

4.2 $d = 1$ (quantum mechanics)

$|\Lambda| = L/a$ sites (discretized time). The Green's function $C(t, t) = 1/(2m_0)$ is **finite** (no UV divergence). The self-energy $\Sigma = O(1/N)$ (no log divergence).

Oscillations at the saddle width: $\sim L/(aN^2)$. For fixed L, a and large N: oscillations vanish. The saddle-point expansion converges for $N > C \cdot L/a$. The mass gap is perturbative.

4.3 $d = 2$ (QFT)

$|\Lambda| = (L/a)^2$ sites. The Green's function $C(x, x) = \frac{1}{4\pi} \log \frac{\Lambda^2}{m_0^2}$ **diverges** logarithmically with the UV cutoff.

The self-energy: $\Sigma \sim \frac{1}{N} \cdot \log \frac{\Lambda}{m_0} \sim \frac{1}{N} \cdot N = O(1)$. The $1/N$ from BL is compensated by the log divergence — **dimensional transmutation**. The gap equation gives $m_0 \sim \Lambda\, e^{-c/g^2}$ (exponentially small).

Oscillations at the saddle width: $\sim |\Lambda|/N^2 = L^2/(a^2N^2)$. For the continuum limit $a \to 0$: oscillations grow, sign problem becomes severe.

4.4 Summary

	$d = 0$	$d = 1$	$d = 2$
$C(x, x)$	$1/m_0^2$	$1/(2m_0)$	$\frac{1}{4\pi} \log \frac{\Lambda}{m_0}$
UV divergence	none	none	**log**
Self-energy Σ	$O(1/N)$	$O(1/N)$	$\mathbf{O(1)}$
Gap equation	trivial	perturbative	**dim. transmutation**
Oscillation at width	$O(1/N^2)$	$O(L/(aN^2))$	$O(L^2/(a^2N^2))$
Sign problem	none	mild	**severe**

The fundamental difference: in $d = 2$, the log divergence of $C(x, x)$ makes the self-energy $O(1)$ instead of $O(1/N)$, producing dimensional transmutation and making the sign problem unavoidable.

5 The saddle-point structure

Despite the difference in operator structure, both models share the same saddle-point analysis.

5.1 The gap equation

On the real σ-axis, the saddle of the exponent in (3) is at $\sigma = 0$ (after absorbing the saddle value into m_0^2 for the NLSM, or into the gap equation for the LSM). The gap equation in both cases:

$$(-\Delta + m_0^2)^{-1}_{xx} = \beta_{\text{eff}} \tag{10}$$

where β_{eff} encodes the coupling. In 2d: $m_0 \sim \Lambda\, e^{-c/g^2}$ (exponentially small).

5.2 The saddle-point Gaussian

Near $\sigma = 0$: the exponent is approximately $-\frac{1}{2}(\sigma, H\sigma)$ where $H > 0$ (positive definite Hessian, including both the bare $1/(2\lambda)$ and the $\operatorname{Tr}\log$ contribution). The saddle-point Gaussian IS log-concave on the real axis.

5.3 The oscillatory corrections

Away from $\sigma = 0$: the exponent develops an imaginary part $\operatorname{Im} f(\sigma) \neq 0$, causing oscillations. The sign problem is the difficulty of controlling these oscillations.

Key scales (at the saddle):

- Width of the Gaussian: $\sigma \sim 1/\sqrt{N}$ per site.
- Oscillation onset: $\operatorname{Im} f \sim O(\sigma^3/\sqrt{N})$ (cubic, since $\operatorname{Im} f'' = 0$ at the saddle).
- Total oscillation at the saddle width: $\sim |\Lambda|/N^2$.

For $N \gg \sqrt{|\Lambda|}$: oscillations are negligible within the saddle-point width. The saddle-point approximation is accurate.

6 Proof strategies

6.1 Strategy A: NLSM (Kupiainen)

Work with the NLSM directly. The constraint gives real mass m_0^2 for free. The FK bound gives $|G| \leq G_{m_0}$ (single-σ decay). Cluster expansion + RP control the σ-average.

Advantage: Real mass in the operator, clean FK bound.
Disadvantage: Cluster expansion needed (technically heavy).
Threshold: $N > C\, m_0^{-a_2}$ with $a_2 = O(d^2)$ (Kupiainen Thm. 1).

6.2 Strategy B: LSM saddle-point

Work with the LSM. No real mass in the operator, but the saddle-point Gaussian on the real σ-axis IS log-concave.

Idea: Apply BL to the saddle-point Gaussian (log-concave, real, positive) and control the oscillatory corrections as a perturbation.

Challenge: The mass gap comes entirely from the σ-average (no single-σ decay). Need to show the oscillatory corrections don't destroy the saddle-point mass.

Advantage: Avoids cluster expansion (BL for the Gaussian).

Disadvantage: No FK bound; need new estimates for the oscillatory σ-average.

6.3 Strategy C: LSM $\to$ NLSM equivalence

Prove that the LSM and NLSM have the same IR physics (they differ only at the σ-mass scale $M_\sigma \sim \sqrt{\lambda}\, v$), then use Strategy A.

Advantage: Gets the NLSM mass gap (with the real mass trick) from the LSM formulation.

Disadvantage: Proving IR equivalence rigorously is itself a major result.

7 The $N = 2$ obstruction

For $N = 2$: the BKT transition occurs at finite λ (Dario-Garban [4]), driven by vortices ($\pi_1(S^1) = \mathbb{Z}$). Any proof must exclude $N = 2$. For $N \geq 3$: $\pi_1(S^{N-1}) = 0$, no BKT.

8 Current status

8.1 What is proved (in Lean)

- HS identity (2): from Mathlib's `fourierIntegral_gaussian`.
- Convergence on the real axis: $|e^{i\sigma a}| = 1$, $|e^{-\sigma^2/(4\lambda)}| \leq 1$.
- HS equivalence (one-site): $\int d\sigma\, e^{-S_{\text{HS}}} = c \cdot e^{-S_{\text{original}}}$ (1 sorry for complex arithmetic).
- Contour rotation algebra: $(i\sigma')^2 = -\sigma'^2$, $i \cdot a \cdot (i\sigma') = -a\sigma'$, etc.
- Continuum limit: OS0–OS2 for the O(N) LSM on T_L^2.

8.2 What is axiomatized

- Contour shift (Cauchy's theorem for rectangles): provable from Mathlib + PNT project.
- Matrix calculus (det C^∞, Jacobi's formula).
- FK sub-Gaussian bound (placeholder for the mass gap step).
- Continuum limit technicalities (4 axioms, porting from pphi2/gaussian-field).

8.3 Open questions

1. Can Strategy B work? Is BL on the saddle-point Gaussian sufficient to control the oscillatory corrections?
2. What is the threshold N_0 for each strategy?
3. Can the N=1 test case ($\mathrm{P}(\varphi)_2$) validate Strategy B?

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